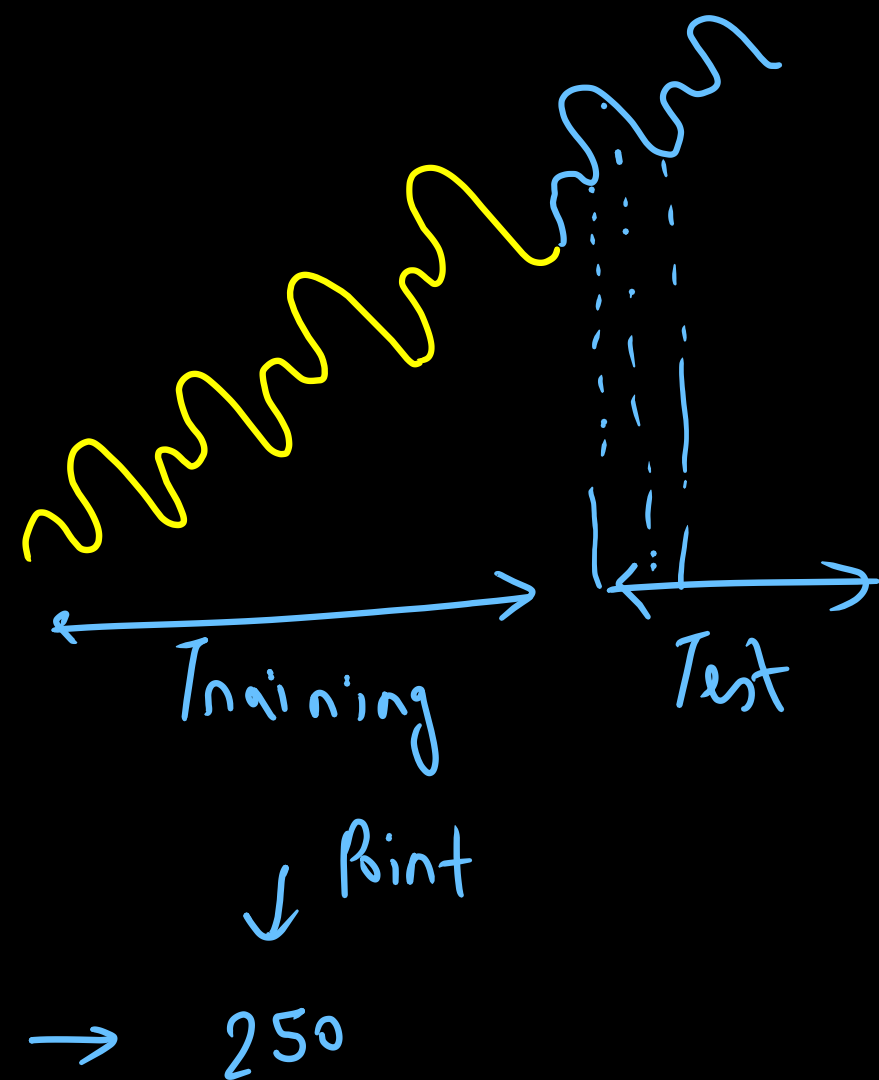


Agenda

→ Confidence Interval

→ Change Point

→ SARIMAX → Exogenous Variable



→ Point forecast

↳ We are predicting exact value.

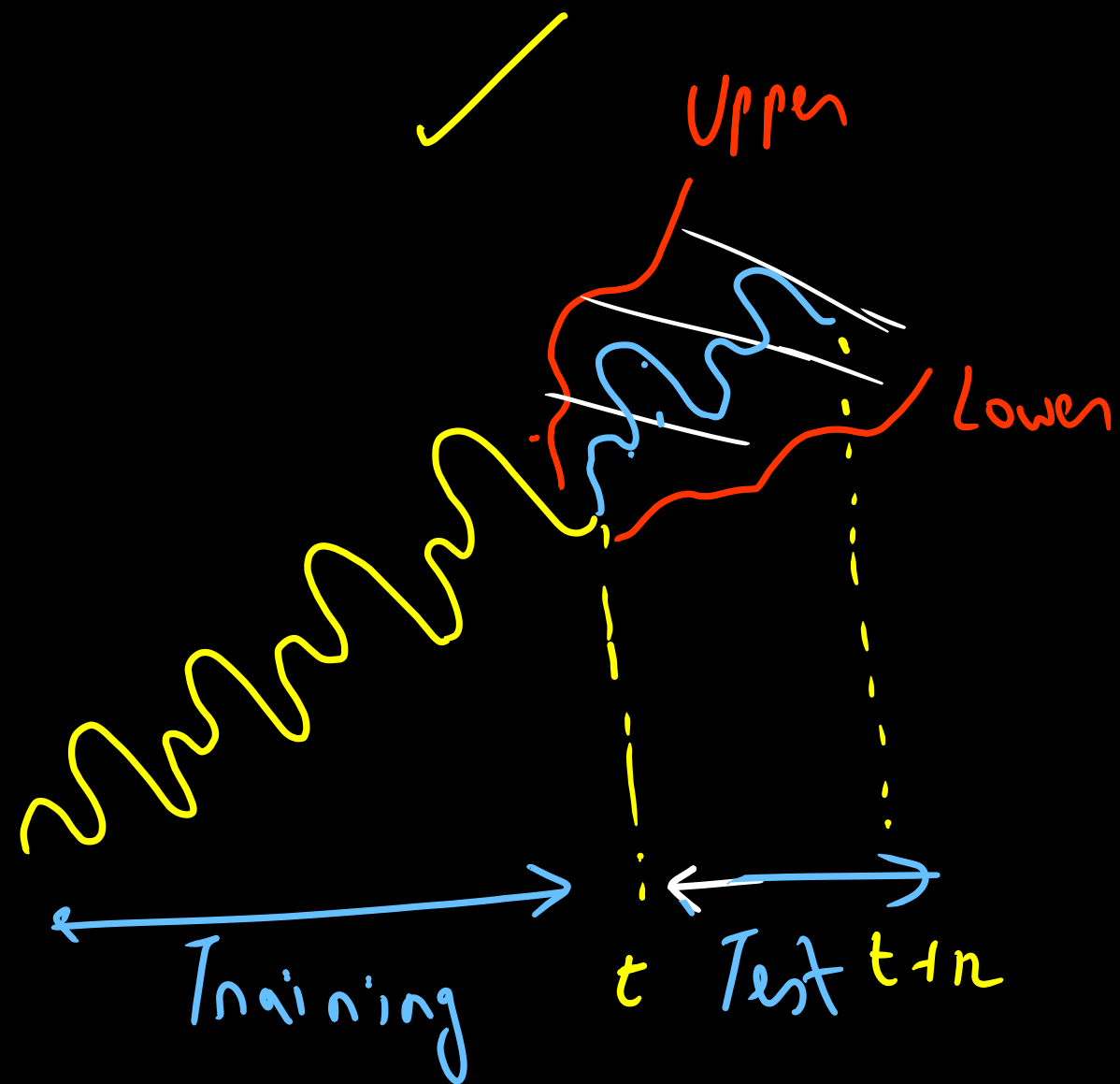
→ Range forecast: We predict upper bound & lower bound.

235 - 265

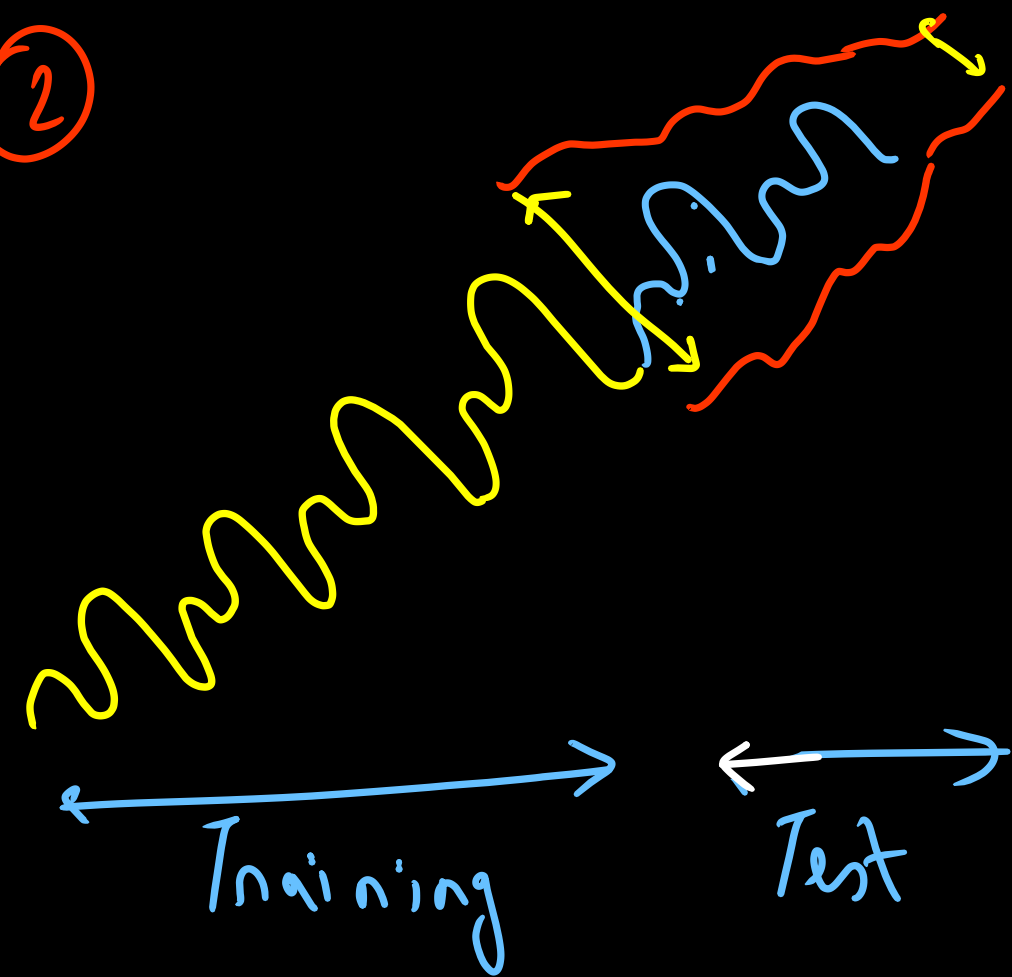
Range forecast

← 250 ± 15

①



②



Confidence Interval

95% $\rightarrow$ 260 - 345

99% $\rightarrow$ 240 - 380

$\rightarrow$ For greater confidence,
wider interval is required

$\rightarrow$ Pratik

✓ 90% confident — 70 — 95 ✓

✓ 100% confidence — 0 — 100

Model A 90%
250 - 255

Model B ✓
90%
252 - 254

↳ Narrow Range

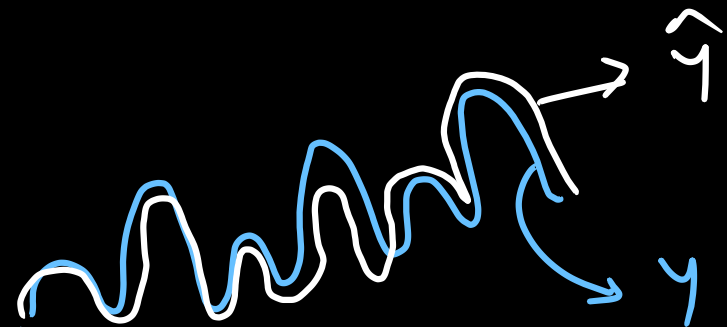
↓

More precise

↓

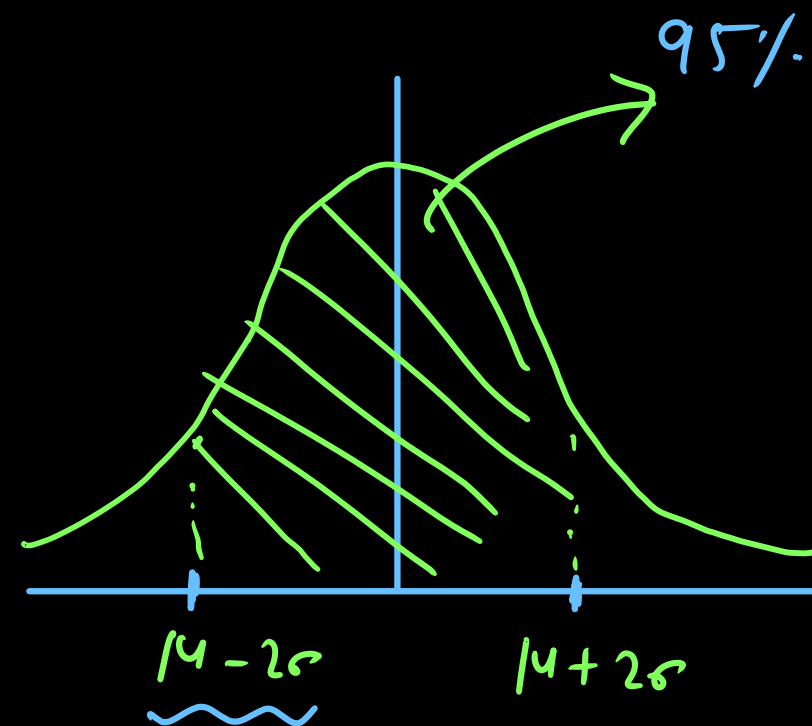
less uncertainty

→ How CI is calculated?



Obtained from
train data

y	$\hat{y}$	$e = y - \hat{y}$
-	-	-
-	-	-
-	-	-



$\mu \pm \sigma \rightarrow 68\%$

$\mu \pm 2\sigma \rightarrow 95\%$

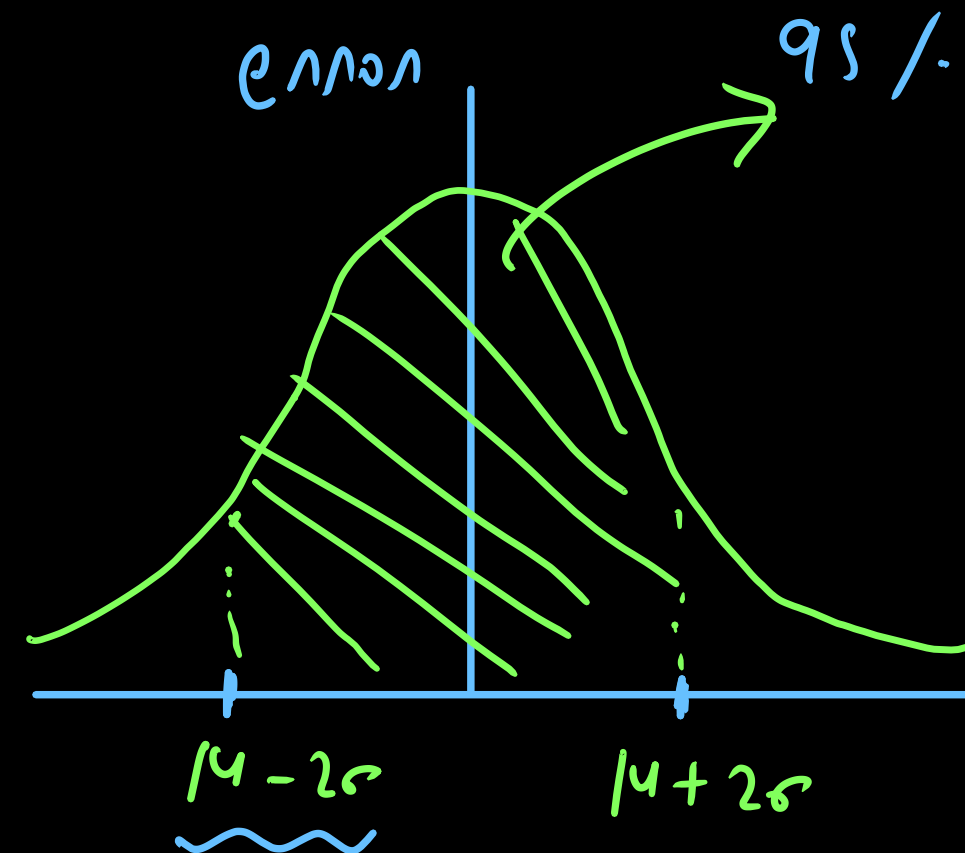
$\mu \pm 3\sigma \rightarrow 99.7\%$

$$\text{Range forecast} = \hat{y}_t + \underbrace{\mu \pm 2\sigma_{\text{residual}}}_{\text{residual}}$$

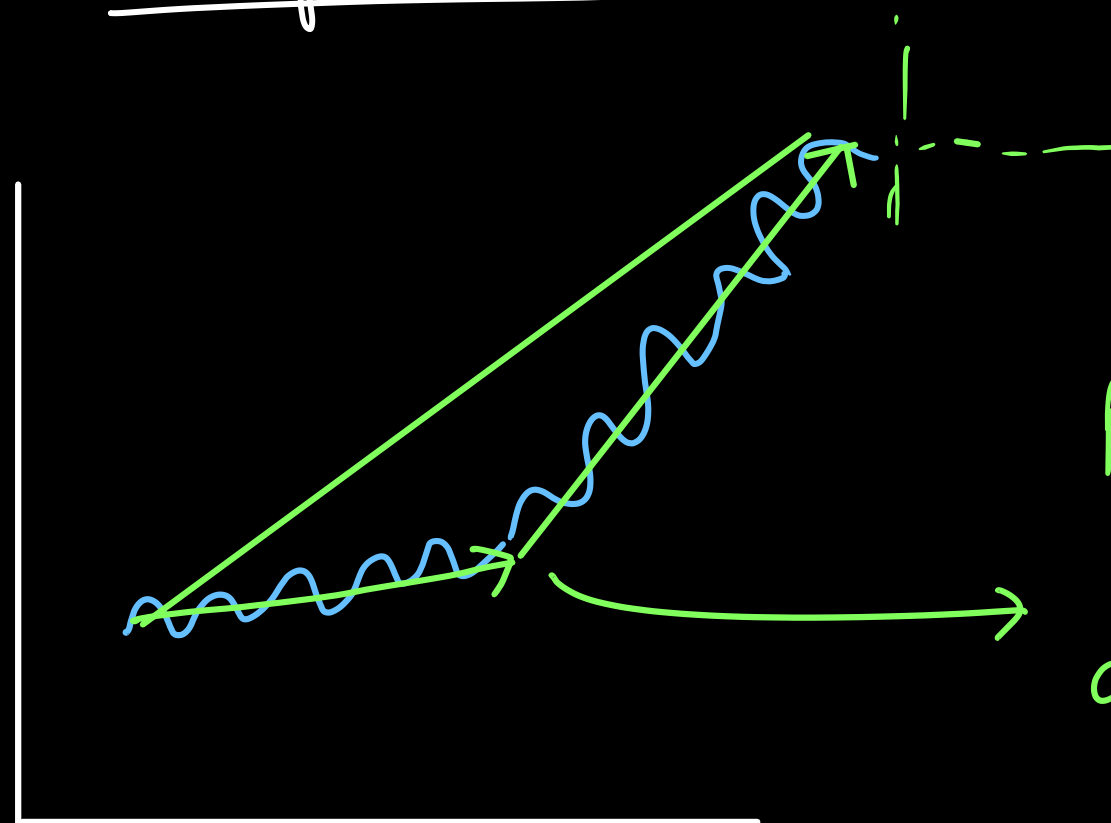
$$\rightarrow \mu_{\text{resi}} = 0 \quad 2\sigma_{\text{resi}} = 7$$

$$\hookrightarrow \text{for CI} = 95\%$$

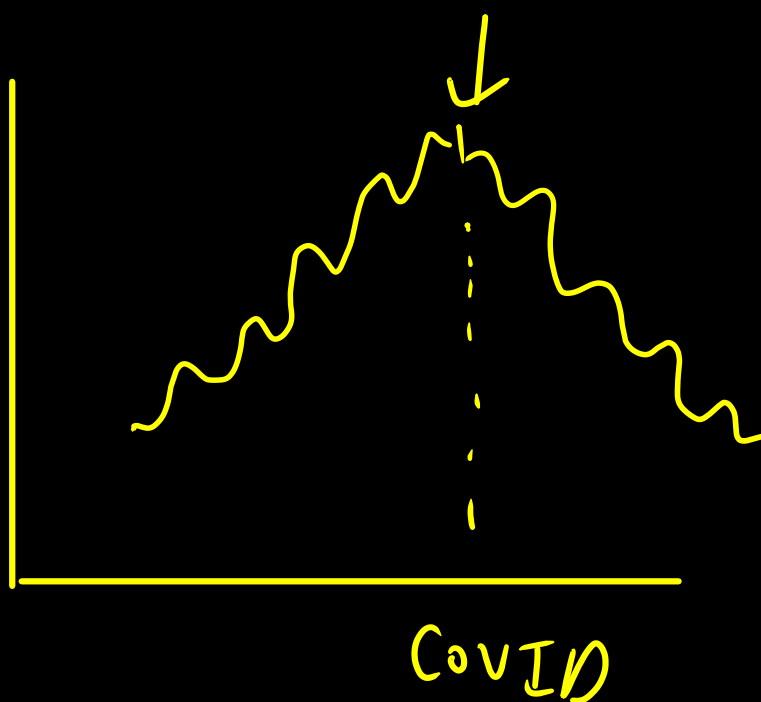
$$\text{Range forecast} = \hat{y}_t \pm 7$$



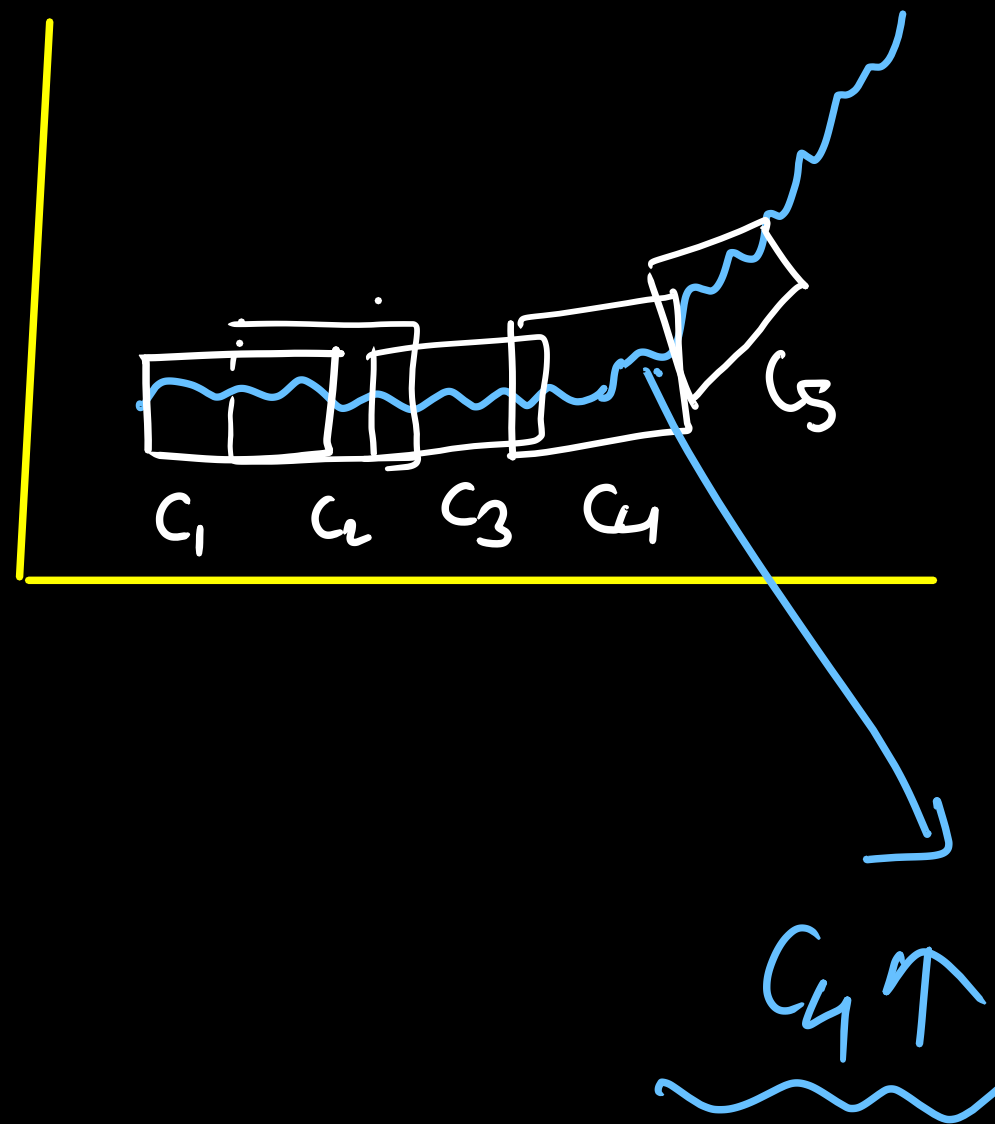
Change Points



Point at which
trend starts
changing



→ Detect change point

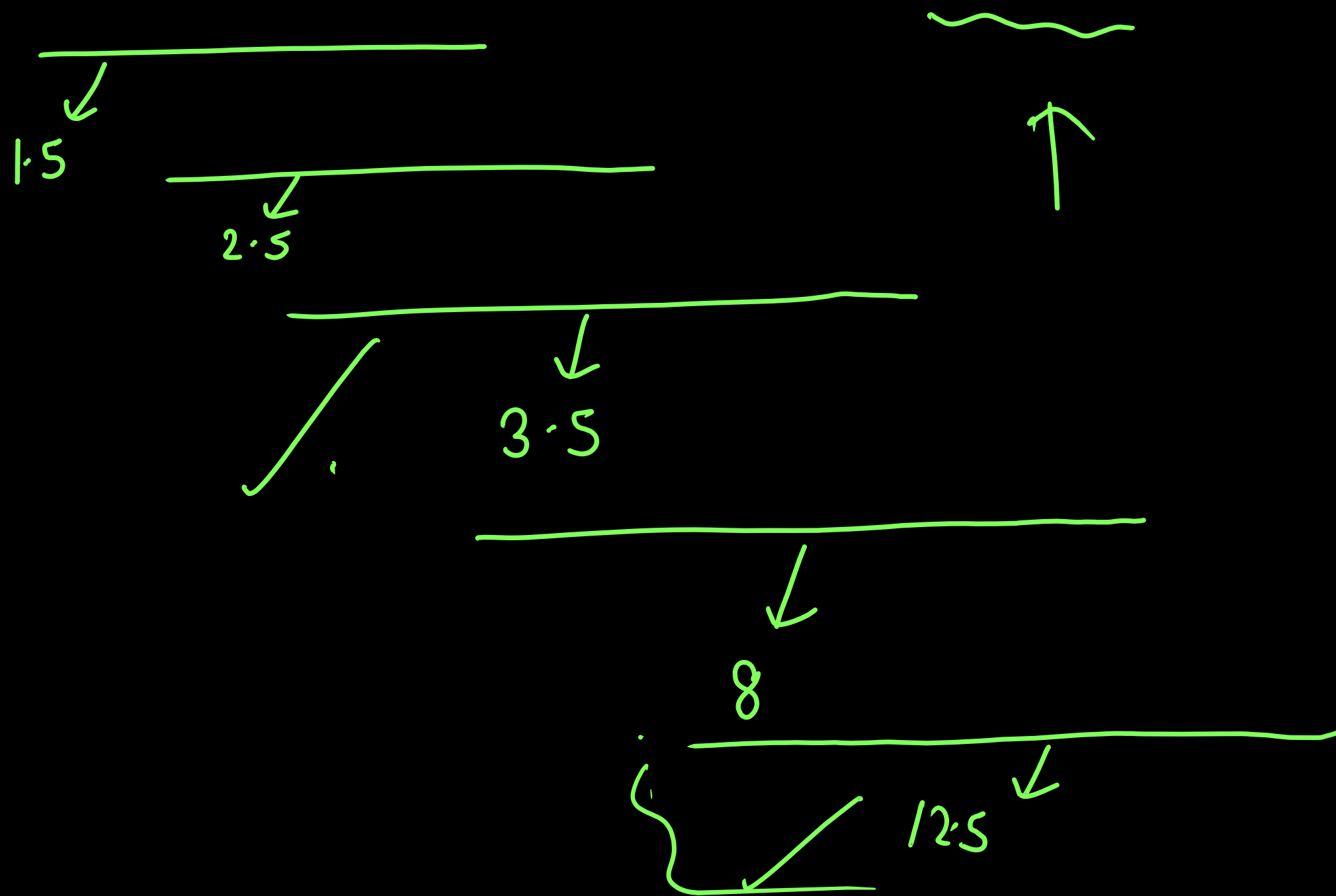


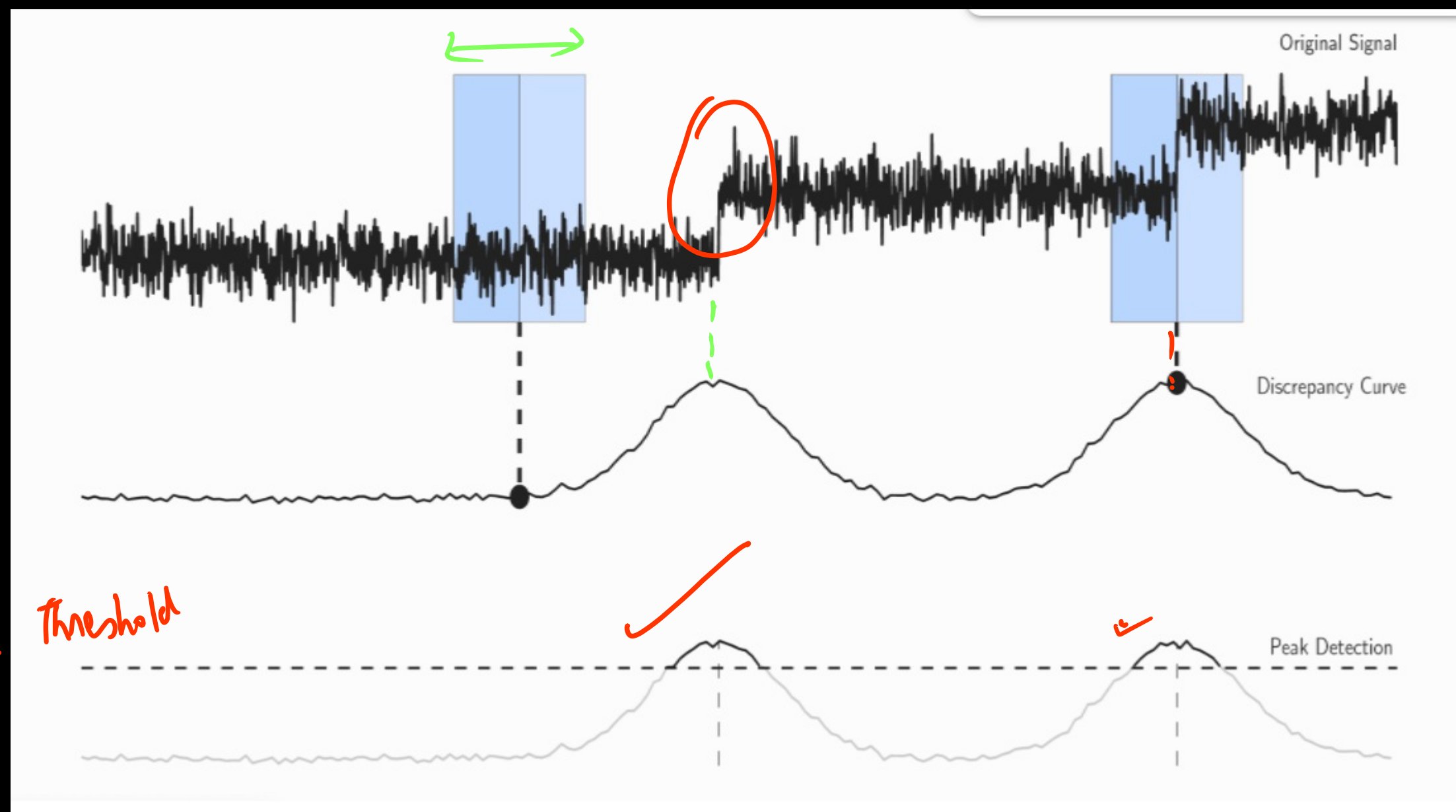
→ Take a window of fixed size & calculate Cost function



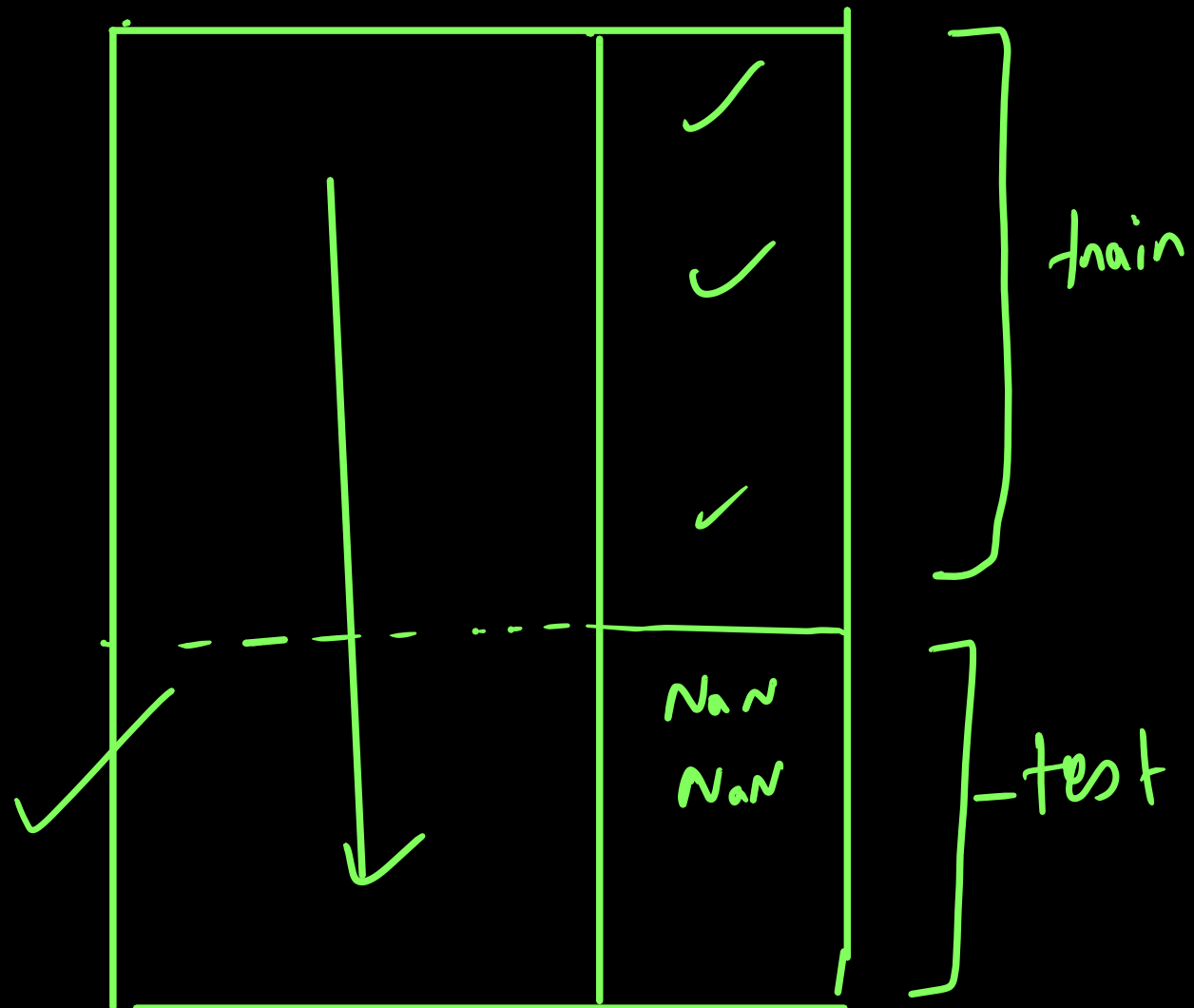
Change in mean,
std, slope

0 1 2 3 4 5 20 21 22 23





total sales



Why is the concept of change points more useful for analysis rather than forecasting?

4 options

Active Duration (Most preferred: 30 seconds)

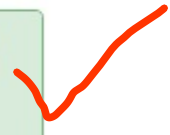
Appears for

20 Secs



A

Change points are unpredictable and can only be confirmed in hindsight.



B

Forecasting models automatically detect and adjust for change points.

C

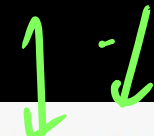
Change points do not affect the accuracy of forecasting models.

D

Forecasting does not require understanding past trends or change points.

```
model = SARIMAX(train['total'], order=(1,0,1), seasonal_order=(1,0,1,7))  
results = model.fit()  
fc = results.forecast(42)
```

AR MA

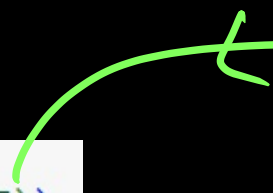


P, D, Q, S

SAR



7



I

P, d, q

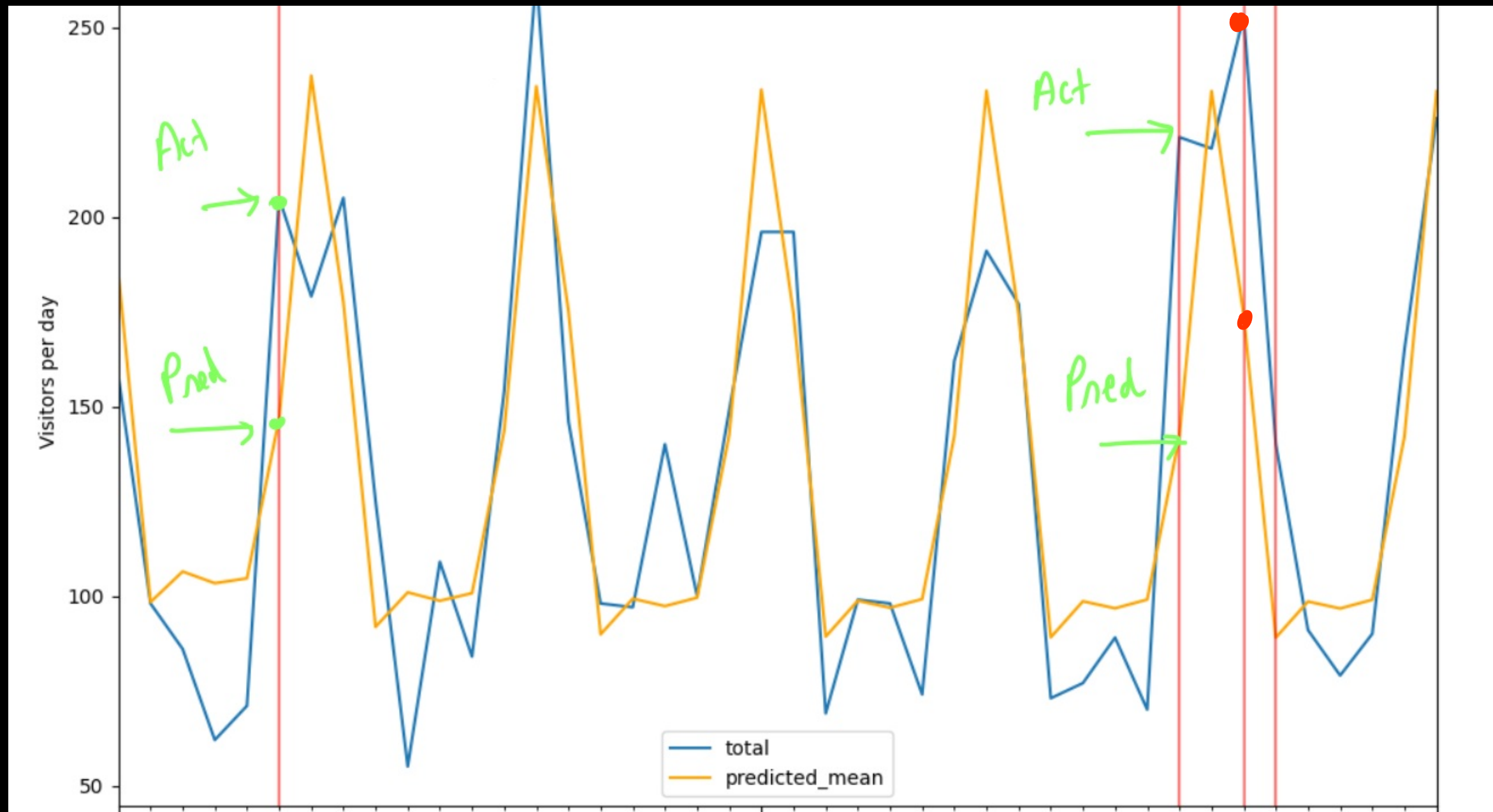


SMA

Season diff



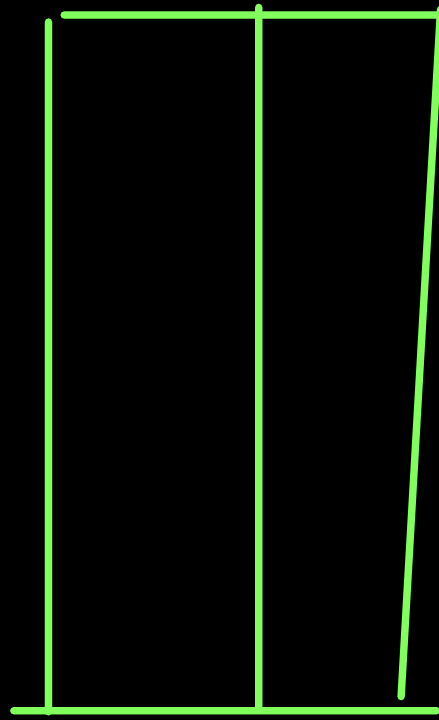
MAPE 0.20



→ Break until 22:23 PM

Exogenous Variable

Datetime y



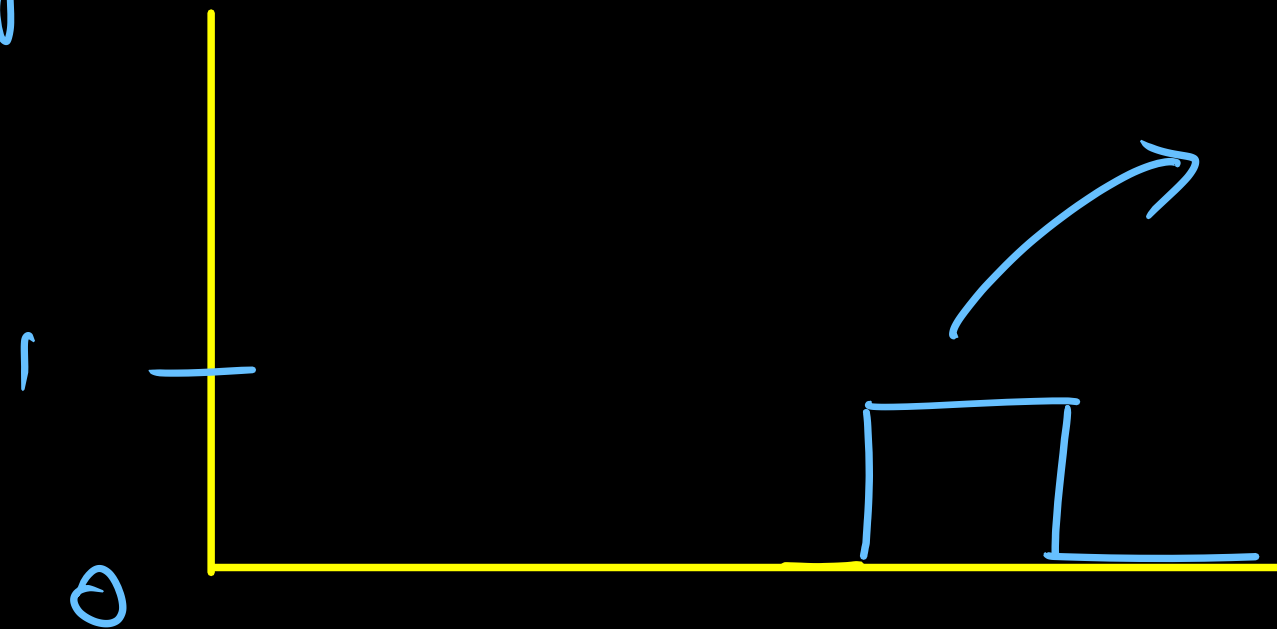
total-sales

```
9] from statsmodels.tsa.statespace.sarimax import SARIMAX  
  
model = SARIMAX(train['total'], order=(1,0,1), seasonal_order=(1,0,1,7))  
results = model.fit()  
fc = results.forecast(42)
```

Exogenous : External variable that impact the Sales(y)

→ Flip Kant

Exogenous



Big billion
days

Holiday : Diwali , Eid

